

# NeurIPS 2026 Workshop Proposal

## 1 Summary

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|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Workshop title          | Real-Time Conversational Agents: Toward Natural Multimodal Interaction                                                                                       |
| Acronym                 | RTCA                                                                                                                                                         |
| Edition                 | 1st                                                                                                                                                          |
| Keywords                | real-time generation; conversational agents; full-duplex speech; talking-head and avatar generation; streaming dialogue; turn-taking; naturalness evaluation |
| Primary contact         | Niki Foteinopoulou ( <a href="mailto:niki@tavus.io">niki@tavus.io</a> )                                                                                      |
| Format                  | Full day, in-person, 7–9h (per NeurIPS 2026 policy)                                                                                                          |
| Audience size           | 50–100 in-person                                                                                                                                             |
| Poster boards requested | 20                                                                                                                                                           |
| Tagline (139/140)       | Real-time conversational agents that listen, see, speak, and act with the naturalness of human interaction — across speech, video, and language.             |
| Location preference     | Sydney                                                                                                                                                       |
| Concurrent proposals    | None. Each organizer is on at most one NeurIPS 2026 workshop proposal.                                                                                       |
| Special requests        | Stable wired (or carved-out Wi-Fi) network for the live-demo session; standard A/V; power at 4–6 demo tables.                                                |

## 2 Topic

Conversational AI has moved from text chat into the real world: voice modes that talk back, embodied avatars that look at us, and agents that share our screens and tools. To feel *natural*, these systems must operate in **real time** — streaming speech, video, and language while continuously listening, watching, and re-planning. This is fundamentally harder than offline generation: latency, partial observability, turn-taking, backchannels, interruptions, and tight cross-modal alignment all become first-class problems that the offline-generation paradigm largely sidesteps. Recent progress on full-duplex audio–language models [5, 7, 23, 25, 8, 16, 15, 9], real-time talking-head and avatar generation [24, 20, 10, 22, 12, 14], low-latency speech synthesis [21, 2, 11], and streaming ASR [17] shows that interactive multimodal agents are now technically feasible. Yet the published record is dominated by *offline* benchmarks, and deployed agents still feel robotic: stilted turn-taking, missing backchannels [6, 19], monotone prosody, awkward interruptions, mis-timed gaze. The community lacks shared vocabulary, benchmarks, and methodology for *interactional* naturalness as opposed to per-utterance quality [1, 3]. RTCA brings together researchers working on this frontier around three intertwined questions: (i) **real-time generation** — producing high-quality speech, video, and language under hard latency budgets, in a streaming or full-duplex fashion; (ii) **naturalness in interaction** — what makes an agent feel like a conversational partner (prosody, gaze, timing, grounding, expressivity, foundational turn-taking dynamics [18]); and (iii) **evaluation of live systems** — how to measure naturalness, responsiveness, and conversational quality where standard offline metrics fall short.

**Why now?** (1) **Real-time is the new frontier and is under-explored.** Full-duplex speech LMs and real-time avatars show the regime is feasible, but methods that work offline (non-causal attention, large beam search, multi-pass refinement, slow diffusion) often do not transfer to streaming. (2) **Naturalness is still missing.** Per-utterance MOS and offline win-rates do not capture turn-taking, interruption, or grounding behaviour. (3) **Fragmentation risk.** Work on these problems is scattered across Interspeech, CVPR/SIGGRAPH, ACL, CHI, and MLSys; without a coordinating venue at NeurIPS the field will be slow to converge on shared problems, datasets, and metrics.

### Topics of interest (CFP scope).

1. Streaming/low-latency speech synthesis, ASR, full-duplex audio LMs [5, 7, 23, 25, 16].
2. Real-time talking-head, avatar, and embodied video generation; lip-sync, gaze, expressivity under streaming [24, 20, 10, 22, 12, 14].
3. Streaming language models; incremental and speculative decoding.
4. Turn-taking, backchanneling, interruption handling [19, 6, 18].
5. Multimodal alignment under latency and partial-observation constraints.
6. Prosody, emotion, and paralinguistic generation in interaction [11, 26].

7. Memory, grounding, tool use during live conversation.
8. Evaluation of naturalness; turn-taking and perceived-latency metrics; interactive Turing-style tests [1, 3, 9].
9. Datasets and benchmarks for *interactive* (not offline) evaluation.
10. Safety, identity, and trust in real-time agents (deep-fakes, persuasion).

### 3 Relationship to Previous Workshops

This is the **1st edition** of RTCA. We have surveyed recent workshops at NeurIPS, ICML, ICLR, ACL, and Interspeech. Closest neighbours and our points of difference:

- **NeurIPS 2025 – MTI-LLM** (long-horizon, text-based multi-turn reasoning). We differ on the *temporal* axis: intra-turn streaming and sub-second latency in multimodal interaction, not long-horizon planning.
- **NeurIPS 2024 – Audio Imagination** (offline audio generation). We differ on *interactivity*: bidirectional, full-duplex, multimodal generation.
- **NeurIPS 2025 – What Makes a Good Video** (offline video generation). We differ on *streaming and conversation*: live audio-driven video under sub-second latency.
- **ML for Audio series**. Broad audio ML, dominated by music and offline modeling. We differ in the focus on interactive, cross-modal, conversational systems.
- **ENLSP series**. Efficiency of language/speech models. We differ in scope: efficiency is a means; our central question is the qualitative experience of interaction.

No recent workshop sits at the intersection of *real-time*, *multimodal*, *conversational*, and *naturalness-evaluated*; RTCA is positioned precisely there, with a live demo showcase and panel that no neighbour workshop has run.

### 4 Invited Speakers

**Alexandre Défossez (Kyutai, Paris)** — *confirmed if Paris; available remotely otherwise*. Researcher and Chief Exploration Officer at Kyutai; lead author of Moshi [5], a real-time full-duplex speech–text foundation model, with prior work on high-fidelity neural audio codecs [4] and source separation. Tentative talk area: design choices and open problems for full-duplex conversational speech models.

**Evonne Ng (Meta Reality Labs / UC Berkeley)** — *provisionally confirmed*. Lead author of *From Audio to Photoreal Embodiment* [14], which generates full-bodied photorealistic conversational avatars from dyadic audio, and of *Body2Hands* [13]. Tentative talk area: synthesising natural body, hand, and face dynamics in interactive conversation.

**Maja Pantic (Imperial College London; FREng)** — *confirmed*. Professor of Affective and Behavioural Computing at Imperial College London and head of the iBUG group, with prior leadership at Meta AI and the Samsung AI Centre Cambridge. Over 350 publications on machine analysis of facial expressions, audio-visual emotion analysis, social-signal processing, and human-centred interfaces; Fellow of the Royal Academy of Engineering. Tentative talk area: behavioural and affective signals as a foundation for naturalness in real-time conversational agents.

**Dimitris Samaras (Stony Brook University)** — *confirmed*. SUNY Empire Innovation Professor and Director of the Computer Vision Lab at Stony Brook, with ~340 publications and ~20K citations spanning computer vision, computer graphics, computational behavioural sciences, gaze prediction, and medical imaging. Tentative talk area: visual modelling of gaze, expression dynamics, and behavioural cues that drive perceived naturalness in interaction.

Per NeurIPS guidance, each invited speaker is chosen for direct topical fit, will confirm in-person attendance at the host geography (with limited remote presentation reserved for the NeurIPS 1-hour allowance), and will confirm their talk is distinct from any other NeurIPS 2026 workshop. Every speaker’s research directly addresses one of the workshop’s pillars.

### 5 Format, Diversity, and Logistics

#### 5.1 Proposed Schedule (single-day in-person, ~8.5h)

A live demo showcase is central; substantial time is allocated to posters and discussion, per the chairs’ guidance.

**08:45–09:00** Welcome.

**09:00–09:40** Invited talk 1 – Real-time speech / full-duplex audio LMs (Défossez; remote at Sydney within NeurIPS 1h allowance).

**09:40–10:20** Invited talk 2 – Streaming video / avatar generation (Ng, provisional).

**10:20–10:45** Coffee + poster setup.

**10:45–11:25** Contributed talks I (4×10 min).  
**11:25–12:05** Invited talk 3 – Topic TBD (Samaras).  
**12:05–13:30** Lunch.  
**13:30–14:30** Poster session with concurrent live-demo booths.  
**14:30–15:10** Invited talk 4 – Topic TBD (Pantic).  
**15:10–15:50** Contributed talks II (4×10 min).  
**15:50–16:15** Coffee.  
**16:15–17:00** **Conversational Agents Showcase** — Demo session: live interactions with deployed / research systems.  
**17:00–17:55** **Panel:** *What does naturalness mean, and how should we measure it?*  
**17:55–18:00** Best paper / best demo announcements; close.

**Discussion mechanisms.** Per-poster seeded discussion questions chosen by area chairs; open “unconference” notepad collects open problems during the day and feeds the closing panel.

## 5.2 Diversity

**Organizers:** the team spans 3 women and 3 men across 2 European countries (UK, Italy) and 4 institutions; affiliations include both industry (Tavus) and academia (King’s College London, Queen Mary University of London, University of Verona); seniority ranges from early-career research scientist to full professor; disciplines span computer vision, machine learning, affective computing, social signal processing, and social robotics / human-agent interaction. **Invited speakers** extend this to 3 countries across 2 continents (France, UK, USA) and explicit academia/industry balance (Kyutai, Imperial College London, Meta/UC Berkeley, Stony Brook), with two women (Ng, Pantic) and two men (Défossez, Samaras) on the current roster. **Program Committee** (TBD): target  $\geq 50\%$  women/non-binary,  $\geq 3$  continents,  $\geq 1$  early-career reviewer per area. **Authors and attendees:** inclusive CFP language; no compute/dataset-size hurdles disadvantaging smaller groups; targeted outreach to LatinX in AI, Black in AI, Women in ML, Queer in AI, and speech communities in the Global South; contributed-talk slots prioritised for student first-authors at comparable quality.

## 5.3 Submission pipeline and timeline

**Tracks:** full papers (8 pages), short papers (4 pages), and demo papers (2 pages, required for the Showcase). Website, CFP, and OpenReview site are pre-prepared and go live on Jul 11.

- **Jul 11** CFP, OpenReview, and website go live; PC outreach begins.
- Aug 8 / Aug 22 – reminders; under-represented-community outreach.
- **Aug 29 (AoE)** submission deadline ( $\sim 7$  weeks from CFP opening).
- Aug 30 – 31 – assignment; COI checks (automated + manual pass by two organizers).
- Sep 1 – Sep 21 – review ( $\sim 3$  weeks; 3 reviews per paper,  $\leq 3$  papers per reviewer; single round, no rebuttal).
- Sep 27 – final decisions; meta-reviews finalised.
- **Sep 29 (AoE) Author notification.**
- **Dec 11 or 12, 2026** workshop day (Sydney).

**COI.** OpenReview tooling plus manual review by two organizers. Per NeurIPS rules: organizers will not submit or give invited talks at RTCA, will not assess submissions from their own institution, and co-authors / advisors / advisees / recent collaborators of any organizer are barred from submitting. Organizers act as the senior reviewing body and may provide supplementary reviews subject to COI; the PC is recruited externally and sized to guarantee 3 reviews per paper. The Niki Foteinopoulou–Ioannis Patras advisor / advisee pair will not be involved in assessing any submission on which either has a conflict; current PhD students / postdocs of either are barred from submitting.

## 5.4 Broader, Social, and Ethical Considerations

Real-time conversational agents now ship in voice assistants, customer support, education, healthcare, and accessibility tools; whether they feel natural materially affects users, including those best served by voice-first or avatar-based interfaces. The workshop sits at a productive intersection with linguistics, conversation analysis, and HCI [18, 19], and we explicitly invite contributions on safety, identity, and trust (impersonation, deepfakes, persuasion). Submissions releasing new datasets must include a brief ethics statement covering consent (voice / likeness), deepfake risk, and provenance. We follow the NeurIPS Code of Ethics, Code of Conduct, and Main Track LLM policy.

## Organizer Information

### 6 List of Organizers

- **Niki Foteinopoulou** :: Research Scientist, Tavus, London, UK. [niki@tavus.io](mailto:niki@tavus.io) [Website] [Scholar]
- **Alessandro Conti** :: Research Scientist, Tavus, Trento, Italy. [Website] [Scholar]
- **Jack Saunders** :: Senior Research Scientist, Tavus, London, UK. [jack@tavus.io](mailto:jack@tavus.io) [Website] [Scholar]
- **Oya Celiktutan** :: Reader in AI and Robotics, King’s College London, UK; Lead of the Social AI & Robotics (SAIR) Laboratory; ELLIS Member. [Website] [Scholar]
- **Cigdem Beyan** :: Associate Professor of Computer Vision and Machine Learning, Department of Computer Science, University of Verona, Italy; ELLIS Member. [Website] [Scholar]
- **Ioannis Patras** :: Professor of Computer Vision and Machine Learning, Queen Mary University of London, UK; Head of the Centre for Multimodal AI; ELLIS Fellow. [Website] [Scholar]

### 7 Organizers’ Experience and Background

**Niki Foteinopoulou** is a Research Scientist at Tavus, working on real-time multimodal conversational agents and the evaluation of conversational naturalness, and a Visiting Lecturer at City, University of London (“Deep Learning for Image Analysis”). She holds a PhD from Queen Mary University of London on multimodal affective computing under uncertainty, with publications at ACM Multimedia, IEEE FG, NeurIPS, and ICCV spanning facial expression recognition, vision–language models, forgery detection, and generative models. At Tavus she works on real-time multimodal conversational agents at scale. She has previously organised a BMVA symposium on AI security.

**Alessandro Conti** is a Research Scientist at Tavus working on multimodal perception, where he focuses on real-time conversational agents. He holds a PhD from the University of Trento on open-world visual recognition with vision–language models, with publications at top-tier computer vision and ML venues.

**Jack Saunders** is a Senior Research Scientist at Tavus working on AI avatars and digital humans, with a research focus on speech-driven facial animation, Gaussian Splatting, video diffusion models, and real-time neural rendering. He holds a PhD from the University of Bath (advisors Vinay Namboodiri and Darren Cosker), with prior research internships at Microsoft and Epic Games, and is the founder of RealSync AI, a digital human consultancy. He previously led the organisation of the BMVA Symposium on Digital Humans and has top-tier publications, including in CVPR, Eurographics and AAIL.

**Oya Celiktutan** is a Reader in AI and Robotics at the Department of Engineering, King’s College London, where she leads the Social AI & Robotics (SAIR) Laboratory. She is also the Honorary Robotics Lead at Guy’s and St Thomas’ NHS Foundation Trust, working closely with two hospitals to translate socially assistive robotic technologies into clinical settings. Her research focuses on multimodal machine learning for autonomous agents that interact seamlessly with humans, spanning multimodal perception, understanding and forecasting human behaviour, and social awareness. She has served as General Chair of ACII 2024, Doctoral Consortium Chair of FG 2024, Publication Chair of ICMI 2025, and Publication Chair of ICSR 2026, and has been an Area Chair at ACM MM, ICMI, and BMVC. She has co-organised the *Expressive Encounters* workshop at ECCV 2024, the *MASSXR* workshop at IEEE VR 2025, the UK Manipulation Workshop 2025, and the *Generation and Evaluation of Non-verbal Behaviour for Embodied Agents* workshop at ACM MM 2025 and ICMI 2026. She is an Associate Editor of the ACM Transactions on Human–Robot Interaction.

**Cigdem Beyan** is an Associate Professor in the Department of Computer Science at the University of Verona, where she co-leads the VIPS (Vision, Images, Patterns and Signals) group and is an affiliated senior researcher in the AI for GOOD lab at Istituto Italiano di Tecnologia. Her research spans computer vision, machine learning, social signal processing, affective computing, and human/animal behaviour analysis. She has served as Area Chair at CVPR (Outstanding Area Chair, CVPR 2026), ECCV, ICRA, BMVC, ACII, and ICMI; Doctoral Consortium Co-Chair of ECCV 2024 and 2026; Demos and Exhibition Chair of FG 2024; Proceedings Co-Chair of AVI 2024; Publicity Co-Chair of IEEE ARSO 2024; and Social Media Co-Chair of RO-MAN 2024. She has co-organised workshops at BMVC, ICPR, and IROS, and led several journal special issues. She is an Associate Editor for IEEE Transactions on Affective Computing, Pattern Recognition, and the ICES Journal of Marine Science, and a member of IEEE, ACM, IAPR, CVPL, and ELLIS.

**Ioannis Patras** is a Professor in Computer Vision and Machine Learning at Queen Mary University of London, Head of the QMUL Centre for Multimodal AI, Director of Research Graduate Studies in the School of Electronic Engineering and Computer Science, and an ELLIS Fellow. His research is at the intersection of computer vision, ML, and NLP, focusing on learning to recognise behaviour, emotions, and cognitive states from images, video, and neuro-physiological signals, and on multimodal learning across vision, language, and audio. He has over 300

publications in venues including CVPR, ICCV, ECCV, NeurIPS, EMNLP, TPAMI, IJCV, and TIP, and leads ~10 PhD students and postdocs at QMUL. He has served as an Area Chair at CVPR, ICCV, FG, ICMI, ACII, and BMVC, and as Associate Editor of Pattern Recognition and Computer Vision and Image Understanding.

## 8 Program Committee

*TBD — target 40–60 reviewers, sized to ensure 3 reviews per paper with no reviewer assigned more than 3 papers, drawn from the speech, vision, language, HCI, and systems communities. Initial outreach begins upon acceptance notification (July 2026). The PC will mirror the diversity commitments above.*

## 9 Concurrent Workshop Involvement

Each organizer is listed on at most one NeurIPS 2026 workshop proposal (this one). No organizer has submitted a related proposal to another concurrent venue.

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